AIRLINE DATA ANALYSIS

Revenue Optimization & Occupancy Rate Study

Using SQL, Python, Pandas & Matplotlib

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Database:	travel.sqlite (8 tables)
Tools Used:	Python | Pandas | SQLite3 | Matplotlib | Seaborn
Scope:	Fleet-wide revenue, booking trends, occupancy rates

1. Business Problem

The airline company operates a diverse fleet of aircraft ranging from small business jets to medium-sized machines, providing high-quality air transportation services to clients for several years. The primary focus is ensuring a safe, comfortable, and convenient journey for passengers. However, the company is currently facing significant challenges due to several compounding external pressures including stricter environmental regulations, higher flight taxes, increased interest rates, rising fuel prices, and a tight labor market resulting in increased labor costs. As a result, the company's profitability is under pressure, and they are seeking data-driven ways to address this issue by conducting a thorough analysis of their database to find opportunities to increase their occupancy rate, which can help boost the average profit earned per seat.

2. Main Challenges

#	Challenge	Description
1	Stricter Environmental Regulations	Growing demand to reduce carbon footprint has resulted in more stringent environmental laws that raise operating costs and restrict expansion potential.
2	Higher Flight Taxes	Governments worldwide are taxing aircraft more heavily to address environmental issues and generate revenue, which raises the cost of flying and decreases passenger demand.
3	Tight Labor Market	The lack of trained personnel in the aviation sector has increased labor costs and increased turnover rates, impacting operational efficiency.

3. Objectives

The primary goal of this analysis is to identify opportunities to increase the occupancy rate on low-performing flights, which can ultimately lead to increased profitability for the airline. Three key strategic objectives guide this study:

Objective	Description
Increase Occupancy Rate	By increasing the occupancy rate, we can boost the average profit earned per seat and mitigate the impact of the challenges facing the business.
Improve Pricing Strategy	Develop a pricing strategy that takes into account changing market conditions and customer preferences to attract and retain customers.
Enhance Customer Experience	Focus on providing a seamless and convenient experience for customers, from booking to arrival, to differentiate the airline in a highly competitive industry and increase customer loyalty.

4. Basic Analysis

The basic analysis of data provides insights into the number of planes with more than 100 seats, how the number of tickets booked and total amount earned changed over time, and the average fare for each aircraft with different fare conditions. These findings will be useful in developing strategies to increase occupancy rates and optimize pricing for each aircraft.

Table 1: Aircraft with More Than 100 Seats

Aircraft Code	Number of Seats
319	116
320	140
321	170
733	130
763	222
773	402

Table 1 - Aircraft models with seat capacity exceeding 100

In order to gain a deeper understanding of the trend of ticket bookings and revenue earned through those bookings, a line chart visualization was utilized. Upon analysis of the chart, the number of tickets booked exhibits a gradual increase from June 22nd to July 7th, followed by a relatively stable pattern from July 8th until August, with a noticeable peak in ticket bookings where the highest number of tickets were booked on a single day. It is important to note that the revenue earned by the company from these bookings is closely tied to the number of tickets booked. Therefore, we can see a similar trend in the total revenue earned by the company throughout the analyzed time period. These findings suggest that further exploration of the factors contributing to the peak in ticket bookings may be beneficial for increasing overall revenue and optimizing operational strategies.

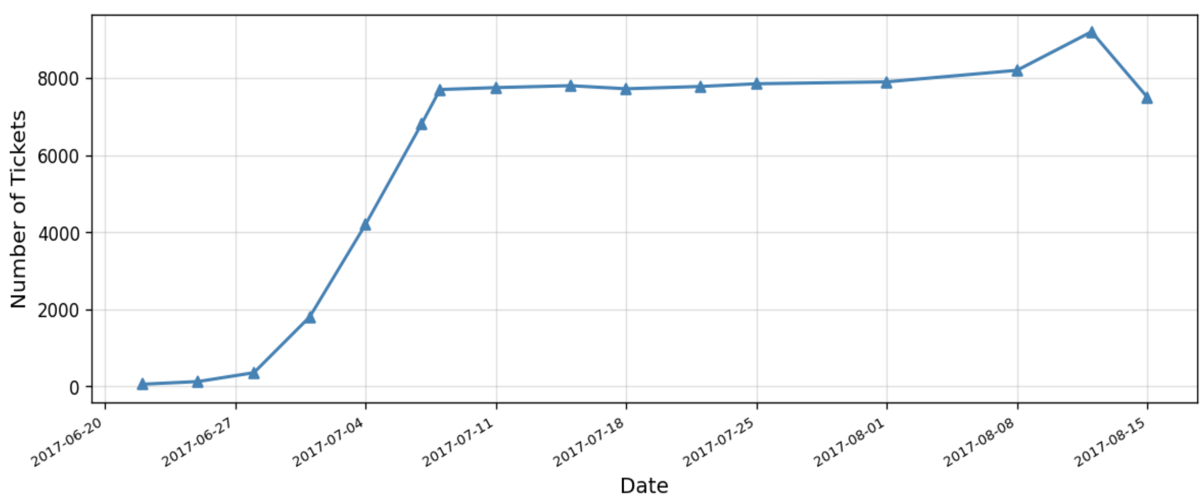

Figure 1 - Number of tickets booked per day (June - August 2017)

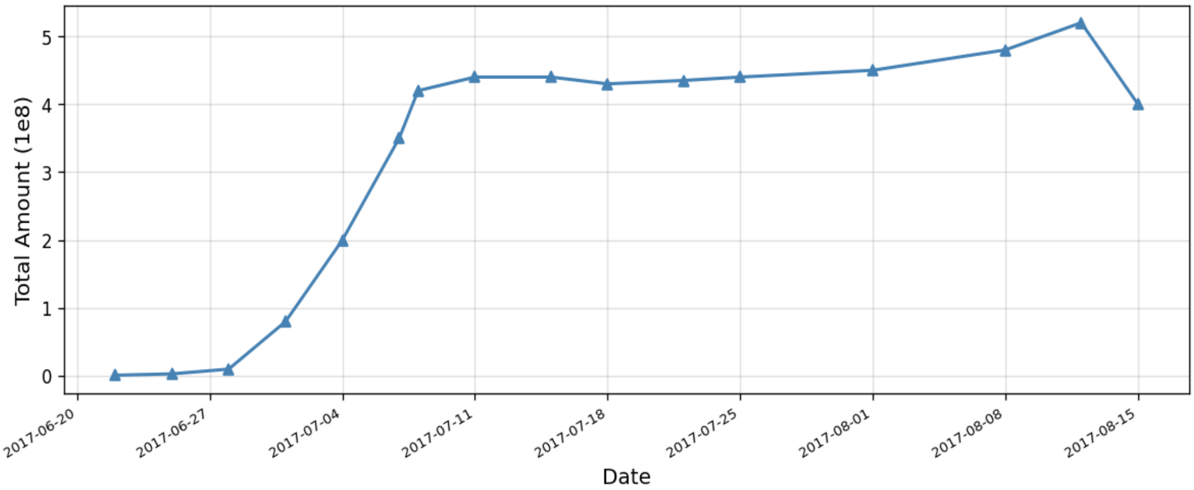

Figure 2 - Total amount earned per day (June - August 2017)

5. Average Charges by Aircraft and Fare Conditions

A bar graph was generated to graphically compare average costs associated with different fare conditions for each aircraft. Figure 3 shows data for three types of fares: Business, Economy, and Comfort. It is worth mentioning that the Comfort class is available on only one aircraft, the 773. The CN1 and CR2 planes only provide Economy class. When different pricing circumstances within each aircraft are compared, the charges for Business class are consistently greater than those for Economy class. This trend may be seen across all planes, regardless of fare conditions.

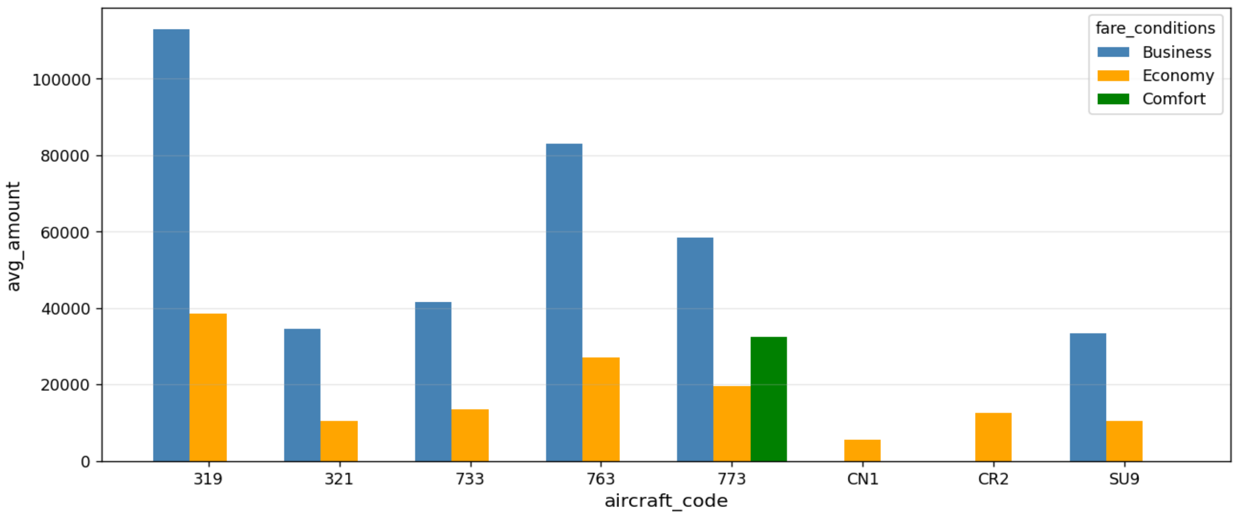

Figure 3 - Average fare amount by aircraft code and fare condition (Business / Economy / Comfort)

6. Analyzing Occupancy Rate

Airlines must thoroughly analyze their revenue streams in order to maximize profitability. The overall income per year and average revenue per ticket for each aircraft are important metrics to consider. Airlines may use this information to determine which aircraft types and itineraries generate the most income and alter their operations appropriately. This research can also assist in identifying potential for pricing optimization and allocating resources to more profitable routes.

The aircraft with the highest total revenue is SU9, and from Figure 3 it can be seen that the price of the Business class and Economy class is the lowest in this aircraft. This can be the reason that most people bought this aircraft ticket as its cost is less compared to others. The aircraft with the least total revenue is CN1, and the possible reason behind this is it only offers Economy class with very low price, possibly due to poor conditions or limited facilities.

Figure 4: Total Revenue, Ticket Count & Average Revenue per Ticket

#	Aircraft Code	Total Revenue	Ticket Count	Avg Revenue / Ticket
0	319	2,706,163,100	52,853	51,201
1	321	1,638,164,100	107,129	15,291
2	733	1,426,552,100	86,102	16,568
3	763	4,371,277,100	124,774	35,033
4	773	3,431,205,500	144,376	23,765
5	CN1	96,373,800	14,672	6,568
6	CR2	1,982,760,500	150,122	13,207
7	SU9	5,114,484,700	365,698	13,985

Figure 4 - Revenue analysis per aircraft type

The average occupancy per aircraft is another critical number to consider. Airlines may measure how successfully they fill their seats and discover chances to boost occupancy rates by using this metric. Higher occupancy rates can help airlines increase revenue and profitability while lowering operational expenses associated with vacant seats. Pricing strategy, airline schedules, and customer satisfaction are all factors that might influence occupancy rates. The occupancy rate is calculated by dividing the booked seats by the total number of seats. A higher occupancy rate means the aircraft seats are more booked and only few seats are left unbooked.

Figure 5: Occupancy Rate per Aircraft

#	Aircraft Code	Booked Seats (Avg)	Total Seats	Occupancy Rate
0	319	53.58	116	0.4619
1	321	88.81	170	0.5224
2	733	80.26	130	0.6173
3	763	113.94	222	0.5132
4	773	264.93	402	0.6590
5	CN1	6.00	12	0.5004
6	CR2	21.48	50	0.4297
7	SU9	56.81	97	0.5857

Figure 5 - Average booked seats vs total seats per aircraft

7. Revenue Impact of 10% Occupancy Rate Increase

Airlines can assess how much their total yearly turnover could improve by providing all aircraft a 10% higher occupancy rate to further examine the possible benefits of raising occupancy rates. This research can assist airlines in determining the financial impact of boosting occupancy rates and whether it is a realistic strategy. Airlines may enhance occupancy rates and revenue while delivering greater value and service to consumers by optimizing pricing tactics and other operational considerations. The table below shows how the total revenue increases after increasing the occupancy rate by 10%, and it gives the result that it will increase gradually, so airlines should be more focused on the pricing strategies.

#	Aircraft Code	Booked Seats	Total Seats	Occupancy Rate	Increased Occ. Rate (+10%)	Projected Annual Turnover
0	319	53.58	116	0.4619	0.5081	2,976,779,410
1	321	88.81	170	0.5224	0.5746	1,801,980,510
2	733	80.26	130	0.6173	0.6790	1,569,207,310
3	763	113.94	222	0.5132	0.5645	4,808,404,810
4	773	264.93	402	0.6590	0.7249	3,774,326,050
5	CN1	6.00	12	0.5004	0.5504	106,011,180
6	CR2	21.48	50	0.4297	0.4726	2,181,036,550
7	SU9	56.81	97	0.5857	0.6443	5,625,933,170

Figure 6 - Projected annual turnover per aircraft with 10% occupancy rate increase

8. Conclusion

To summarize, analyzing revenue data such as total revenue per year, average revenue per ticket, and average occupancy per aircraft is critical for airlines seeking to maximize profitability. Airlines can find areas for improvement and modify their pricing and route plans as a result of assessing these indicators. A greater occupancy rate is one important feature that can enhance profitability since it allows airlines to maximize revenue while minimizing costs associated with vacant seats. The airline should revise the price for each aircraft as both excessively low and high pricing are factors that prevent people from buying tickets on those aircrafts. The airline should decide a reasonable price according to the condition and facility of the aircraft, which should not be at either extreme.

Furthermore, boosting occupancy rates should not come at the price of consumer happiness or safety. Airlines must strike a balance between the necessity for profit and the significance of delivering high-quality service and upholding safety regulations. Airlines may achieve long-term success in a highly competitive business by adopting a data-driven strategy to revenue analysis and optimisation.

Key Takeaways
Aircraft 773 has the highest occupancy rate at 65.9%, making it the most efficiently utilized aircraft in the fleet.
Aircraft CR2 has the lowest occupancy rate at 42.97%, representing the biggest opportunity for improvement.
Aircraft SU9 generates the highest total revenue (5.1B) due to its low pricing attracting the highest ticket volume (365,6
A 10% across-the-board occupancy rate increase would generate the highest absolute gain for aircraft SU9 (+511M) a
Pricing strategy is the most actionable lever: business-class fares are consistently higher, yet occupancy gaps remain,